

AI-assisted extraction of architectural parameters from RISC-V specifications

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Problem Statement

Link to actual assignment :

https://docs.google.com/document/d/1IbOHtC-BbKbRd_3a6rB_8158lF9iR7Pi/edit

AI-assisted extraction of architectural parameters from RISC-V specifications

Develop prompts which extract the parameters from the snippets of text from the ISA Manual. Usually, usage of following words implies a parameter: "may/might/should," "optional/optionally" and "implementation defined" or "implementation specific".

Please submit the following information:

1. Details about the LLM(s) used (name, version, context length etc.)
2. Prompts: Especially how you developed and refined the prompt, dealt with model hallucinations etc.
3. Results formatted as a YAML file with fields for name, description, type and constraints etc.

Snippets:

Privileged Spec 19.3.1:

Caches organize copies of data into cache blocks, each of which represents a contiguous, naturally aligned power-of-two (or NAPOT) range of memory locations. A cache block is identified by any of the physical addresses corresponding to the underlying memory locations. **The capacity and organization of a cache and the size of a cache block are both implementation-specific**, and the execution environment provides software a means to discover information about the caches and cache blocks in a system. In the initial set of CMO extensions, the size of a cache block shall be uniform throughout the system.

Privileged Spec 2.1:

"Conventional" R/W accessibility of CSRs according to [address mapping](#)

The standard RISC-V ISA sets aside a 12-bit encoding space (csr[11:0]) for up to 4,096 CSRs. By convention, the upper 4 bits of the CSR address (csr[11:8]) are used to encode the read and write accessibility of the CSRs according to privilege level as shown in [Table 1](#). The top two bits (csr[11:10]) indicate whether the register is read/write (00,01, or 10) or read-only (11). The next two bits (csr[9:8]) encode the lowest privilege level that can access the CSR.

Master Prompt

You are an expert at reading RISC-V specifications and documentations. Your job is to extract parameters from snippets of texts.

<Context on Input>

- You will be given snippets of text that are based out of ISA
- ISA stands for Instruction Set Architecture. It is the official specification that defines how a RISC-V processor must operate, including its instructions, registers, and optional features.

<Context on Output>

- Your job is to extract parameters from the input ISA manual snippet.
- The Output should strictly follow the below format.
- No additional text must be provided that does not follow the output YAML guidelines
- Output must be generated for each and every parameter. I.e if there are 2 parameters in input snippet, then output should have ONLY 1 YAML file
- Each parameter should follow output guidelines.

<Output Guidelines>

- Output should be a YAML file
- Output for each parameter should STRICTLY have the following fields

1) Name : A way to uniquely identify parameter (IF no specific name provided then create a name that can uniquely identify the parameter)

2) Type : Data type (Example : string , boolean etc , follow C datatypes)

3) Constrains : (in lexicographic order if multiple exists)

4) Description : Everything about this parameter that does not fall in above categories

- NOTE : if a field information is not available, fill it with "NA"

<How to Identify Parameters>

Use of the following words imply existence of Parameters

- "May / might / should,"
- "Optional / optionally"
- "implementation defined" / "implementation specific".

Outputs

- **Model Name:** GPT-3.5
- **Provider:** OpenAI
- **Interface:** ChatGPT (Free Web Version)

- **Context Length:** ~16K tokens (approximate upper bound for GPT-3.5-turbo variants)

Snippet 1 : Input

Privileged Spec 19.3.1:

Caches organize copies of data into cache blocks, each of which represents a contiguous, naturally aligned power-of-two (or NAPOT) range of memory locations. A cache block is identified by any of the physical addresses corresponding to the underlying memory locations. The capacity and organization of a cache and the size of a cache block are both implementation-specific, and the execution environment provides software a means to discover information about the caches and cache blocks in a system. In the initial set of CMO extensions, the size of a cache block shall be uniform throughout the system.

Output

- Name: CacheCapacity

Type: size_t

Constrains: implementation-specific

Description: The total storage capacity of a cache. The capacity is not architecturally fixed and may vary between implementations. Software may discover this information through mechanisms provided by the execution environment.

- Name: CacheOrganization

Type: string

Constrains: implementation-specific

Description: The internal structural organization of the cache, such as how cache blocks are arranged and managed. This property is defined by the implementation and is not fixed by the ISA.

- Name: CacheBlockSize

Type: size_t

Constrains: implementation-specific

Description: The size of an individual cache block, representing a contiguous, naturally aligned power-of-two (NAPOT) range of memory locations. This size may vary across implementations, but software can discover it through execution-environment mechanisms.

- Name: UniformCacheBlockSize

Type: boolean

Constrains: shall be uniform throughout the system

Description: Indicates that, in the initial set of CMO extensions, the cache block size must be the same across all caches in the system.

Snippet 2 : Input

Privileged Spec 2.1:

"Conventional" R/W accessibility of CSRs according to address mapping The standard RISC-V ISA sets aside a 12-bit encoding space (csr[11:0]) for up to 4,096 CSRs. By convention, the upper 4 bits of the CSR address (csr[11:8]) are used to encode the read and write accessibility of the CSRs according to privilege level as shown in Table 1. The top two bits (csr[11:10]) indicate whether the register is read/write (00,01, or 10) or read-only (11). The next two bits (csr[9:8]) encode the lowest privilege level that can access the CSR.

Output :

- Name: CSRAddressWidth
Type: uint32_t
Constrains: 12-bit
Description: The standard RISC-V ISA reserves a 12-bit encoding space (csr[11:0]) for CSR addresses.

- Name: CSRMaximumCount
Type: uint32_t
Constrains: up to 4096
Description: The total number of CSRs addressable within the 12-bit CSR encoding space.

- Name: CSRReadWriteAccessibilityEncoding
Type: uint8_t
Constrains: csr[11:10]=00,01,10 for read/write; csr[11:10]=11 for read-only
Description: The top two bits of the CSR address encode whether a CSR is read/write or read-only.

- Name: CSRPrivilegeLevelEncoding
Type: uint8_t
Constrains: csr[9:8] encode lowest privilege level
Description: The next two bits of the CSR address specify the lowest privilege level permitted to access the CSR.